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A systematic review on the impact of healthcare consolidation in the digital era

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ABSTRACT

Objectives: This systematic review examines how healthcare provider consolidation, particularly among hospitals and health systems, affects Health Information Technology (HIT), with a focus on Electronic Health Records (EHRs) and interoperability.

Methods: Following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, we reviewed literature published through June 2023 using major databases such as PubMed and Cochrane. Data from 16 peer-reviewed studies covering 125 quantitative analyses were synthesized to analyze how provider consolidation shapes HIT infrastructure and performance, including information exchange, EHR vendor diversity, and downstream outcomes such as healthcare access, cost efficiency, and care quality.

Results: The findings show that consolidation is significantly associated with improvements in information exchange and increased healthcare utilization, particularly in emergency department and primary care settings. While the improvements are encouraging, they are often incremental and do not yet reflect full interoperability, which involves not only data sharing but also the seamless, meaningful use of information across systems. Evidence on cost-effectiveness, care quality, and patient outcomes is mixed.

Conclusions: Consolidation may improve certain aspects of digital infrastructure, especially data exchange, but does not guarantee full interoperability or better downstream outcomes. These findings point to the need for future research to go beyond measuring data exchange by assessing actual interoperability performance and examining the real-world challenges of HIT integration in consolidated systems. As healthcare consolidation continues, careful evaluation of its digital, clinical, and organizational impacts will help advance interoperability and support equitable, high-quality care.

Public Interest Summary: This review explores how hospital and health system consolidations affect Health Information Technology (HIT), particularly Electronic Health Records (EHRs) and interoperability. Through a systematic search, we identified 16 papers and 125 quantitative analyses. The findings show that consolidation is significantly associated with improved information exchange and increased healthcare use. While the improvements are encouraging, they are often partial and do not mean that systems can fully and effectively use shared data across all settings. The impact on cost-effectiveness, care quality, and patient outcomes varies across studies. Our review highlights the need for more research that looks beyond basic data sharing to evaluate how well health systems actually use information and to understand the practical challenges of integrating technology after hospitals or health systems merge. The findings also offer insights to inform policy efforts aimed at promoting equitable, high-quality care in an increasingly digitally integrated healthcare environment.

Introduction

Since the passage of the Affordable Care Act (ACA), healthcare provider consolidation, particularly among hospitals and health systems, has accelerated markedly [1–3]. Between 2010 and 2019, around 1500 hospitals were involved in mergers or acquisitions, reshaping the

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